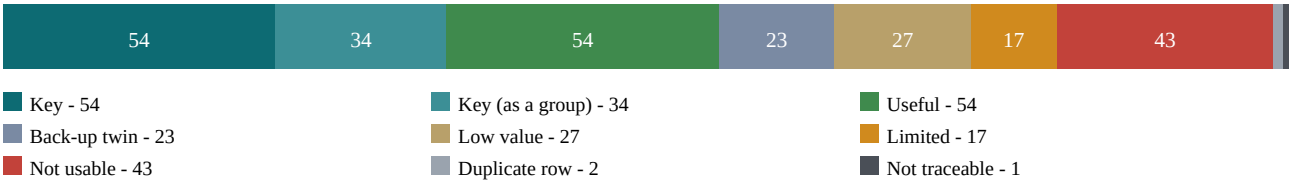
SOFC PoC · sensor list check · 15 Oct 2026

Can we trace every sensor in the list, and which ones are worth using?

254 of the 255 rows can be traced to data the plant actually logged. Only **LIR 7,01** never reached the logger. About half the list carries the information we need. Roughly one in six rows (mostly 40 stack package-voltage channels) records nothing usable.

Checked: *sensors displayed in the visualization tool.xlsx* as saved 15 Oct 2026 11:51 (255 rows), against 1-minute averages of all 39 logged days (30 Oct 2025 – 30 Mar 2026) and the plant stages.



Contents: 1 Overview · 2 Can we find each tag? · 3 Which sensors are useful? · 4 Problems and gaps · 5 All 255 rows

Verdict	Meaning	Rows
Key	Tells us what the plant is doing; stage rules, safety trips or power output depend on it.	54
Key (as a group)	34 working stack thermocouples: use together as mean, hottest spot and spread.	34
Useful	Works and changes clearly from stage to stage.	54
Back-up twin	Moves 1:1 with a neighbour; keep for safety, one is enough for models.	23
Low value	Near room temperature / supply pressure whatever the plant does, or a setting.	27
Limited	Real readings for only part of the 39 days.	17
Not usable	Stuck, empty or clearly in the wrong place.	43
Duplicate row	Same tag twice in the list.	2
Not traceable	No data column exists for this tag.	1

If you remember one thing: Almost the whole list (254 of 255) is in the data. The real issue is the 43 rows that record nothing usable.

2 · Can we find each tag in the logged data?

Yes for 254 of 255. Rule for 199 tags: the logger writes the tag with underscores (TIR 4,6,12 → TIR_4_6_12); flow controllers use the actual-value column (FIRC 1,1,1 → ct_FIRC_111_PV_ProcessValue).

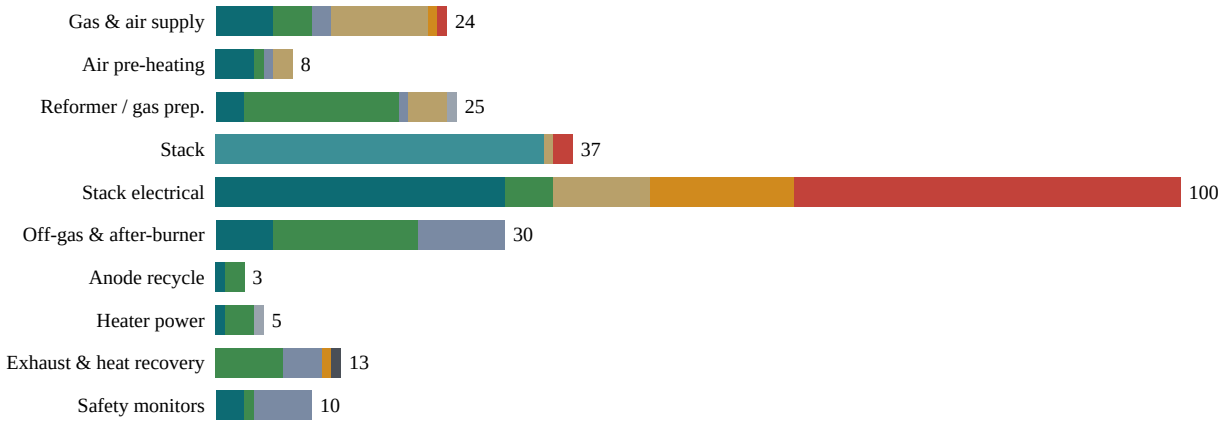
How found	Rows	Examples / note
Same name	199	All TIR/TIRC/TIRZ, PIR/PIRZ, EIR, most FIRC
Other name	30	TIRC 3,19,x → TIR_3_19_x; QIR 1,4 → PLC_pi_QIR_1_4_ActVal_Value; EIRC 2,01 → EIRC_2_01_ActV_In_ThreePhasePower; EIRC 4,s,1/2/4/5/6 → PLC_hostSetp_ctrl_ctEIRC4s_… (all 39 days; the EIRC_4s_… look-alikes start 17 Mar and are shifted on 20 Mar)
Likely match	8	EIRC 3,1,1 / 5,1,1 / 5,1,2 → heater currents pi_IIR_…_ActVal (controller output itself not logged); EIRC 4,s,9 → …rVoltCellMin_ActV
Partly	17	FIRC 1,1,3 from 17 Mar 2026 (tag renamed); PIR 7,1,2 reads −1 until 17 Feb 2026; EIRC 4,s,3/7/8 from 17 Mar 2026
Not found	1	LIR 7,01: no column among the 836 logged columns

The ICA instrument list was not on this computer; EIRC 4,s,3 / 8 / 9 still need to be confirmed against it.

If you remember one thing: Use the PLC copy (PLC_hostSetp_ctrl_ctEIRC4s_…) for string load values; it covers all 39 days.

3 · Which sensors carry useful information?

The hot parts of the plant carry most of the information: stack, heaters, reformer and after-burner. Supply lines and the package-voltage channels add very little.



Rows per plant area, coloured by verdict (same colours as page 1).

Judged on four points: (1) alive: share of real minutes and days it moves; (2) follows the plant: share of its variation explained by the stage (about 60 % for stack/heater temperatures, mostly under 20 % for supply pressures); (3) a copy: rank correlation ≥ 0.96 with a sensor at the same spot; (4) already used by the stage rules or trip checks.

Key (54)

Flows that define each phase, EAPH heater, heater/reformer temperatures, safety trips, string voltages and currents.

FIRC 1,1,1, FIRC 1,1,2, PIRZ 8,1,2, FIRC 1,4, FIRC 1,2, FIRC 1,3, FIRC 1,5, TIR 2,05, TIR 2,03, EIRC 2,01, TIRZ 8,1,10, TIRC 5,17, TIRC 6,1,1, TIRC 5,3,1, TIRC 5,3,2, TIRC 5,09, TIRC 3,07, TIRC 3,15, TIRC 3,1,1, PIR 2,02, TIRC 2,06, TIRZ 8,1,3, TIRC 5,1,1, EIR 4,1,12, EIR 4,1,15, EIR 4,1,18, EIR 4,2,12, EIR 4,2,15, EIR 4,2,18, EIR 4,3,12, EIR 4,3,15, EIR 4,3,18, EIR 4,4,12, EIR 4,4,15, EIR 4,4,18, EIR 4,5,12, EIR 4,5,15, EIR 4,5,18, EIRC 4,1,1, EIRC 4,1,2, EIRC 4,1,9, EIRC 4,2,1, EIRC 4,2,2, EIRC 4,2,9, EIRC 4,3,1, EIRC 4,3,2, EIRC 4,3,9, EIRC 4,4,1, EIRC 4,4,2, EIRC 4,4,9, EIRC 4,5,1, EIRC 4,5,2, EIRC 4,5,9, TIRC 5,2,1

Useful (54)

After-burner profile, reformer bed profile, off-gas pressures, exhaust temperatures, heater currents.

PIR 1,1,2, PIR 1,4,2, PIR 1,2,2, PIR 1,3,2, PIR 2,01, TIR 7,04, PIR 7,02, PIR 7,03, PIR 7,1,1, TIR 7,2,1, TIR 5,6,1, TIR 7,1,1, TIRC 3,17, TIR 7,3,1, TIR 3,06, TIRC 3,03, PIR 3,02, PIR 3,01, TIR 6,01, PIR 6,01, TIR 5,4,1, PIR 5,3,1, TIRC 5,5,1, PIR 5,4,1, EIRC 5,1,1, EIRC 5,1,2, TIRZ 8,1,9, TIRC 5,07, TIRC 5,08, TIRC 5,10, TIRC 5,11, TIRC 5,12, TIRC 5,13, TIRC 5,14, TIRC 5,15, TIRC 3,08, TIRC 3,09, TIRC 3,10, TIRC 3,11, TIRC 3,12, TIRC 3,13, TIRC 3,14, TIRC 3,16, EIRC 3,1,1, PIR 3,03, TIR 3,18, TIRC 3,19,1, PIR 5,1,1, EIRC 4,1,4, EIRC 4,2,4, EIRC 4,3,4, EIRC 4,4,4, EIRC 4,5,4, PIR 5,2,1

Back-up twin (23)

Moves with a partner (control/safety pair or line 1/line 2).

PIR 1,1,3, TIR 1,4,2, TIRZ 8,1,1, TIR 7,05, TIR 7,3,2, TIR 2,1,2, TIR 7,2,2, TIR 7,1,2, TIR 5,6,2, TIRZ 8,2,5, TIRZ 8,2,4, TIR 5,4,2, PIR 5,3,2, TIRC 5,5,2, PIR 5,4,2, TIRZ 8,2,2, TIRZ 8,2,3, TIRZ 8,2,1, TIRC 3,19,2, TIRC 5,1,2, PIR 5,1,2, TIRC 5,2,2, PIR 5,2,2

Low value (27)

Gas-inlet temperatures, constant supply pressures, load settings.

PIR 1,1,1, TIR 1,1,1, PIR 1,4,1, TIR 1,4,1, PIR 1,2,1, PIR 1,5, PIR 1,3,1, TIR 1,2,1, TIR 1,2,2, TIR 1,3, TIR 3,01, TIR 2,04, TIR 3,04, TIR 2,1,1, TIR 3,05, TIR 3,02, TIR 4,1, EIRC 4,1,5, EIRC 4,1,6, EIRC 4,2,5, EIRC 4,2,6, EIRC 4,3,5, EIRC 4,3,6, EIRC 4,4,5, EIRC 4,4,6, EIRC 4,5,5, EIRC 4,5,6

Key (as a group) (34)

TIR 4,6,1 … 4,6,36 except 4,6,19 and 4,6,25.

If you remember one thing: About 140 signals do the job for stage and synthetic-data work; twins and low-value rows can stay on screen but out of models.

4 · Problems and gaps

Problem	Rows	What we see
Package voltages EIR 4,s,10/11/13/14/16/17/19/20	40	Real ≈5 V only 13–15 Nov 2025; ≈0.005 V since 16 Nov (4,4,16/17 never real). Only stack totals .12/.15/.18 work.
TIR 4,6,19	1	Stuck at 1300 °C on all days: broken thermocouple.
TIR 4,6,25	1	Reads 2–66 °C while neighbours read 500–650 °C: not in the stack.
QIR 1,4	1	≈0 on nearly all days, four short spikes.
Duplicate rows	2	Row 64 EIRC 2,01 (red) repeats row 29; row 55 TIRC 3,17 repeats row 45.
Limited	17	Usable only in the periods named on page 2.

Logged and useful, but not in the list

Signal	Logged column	Why	Advice
EIR 4,s,21 (5)	pi_EIR_4_s_21	Stack-4 voltage of every string; list stops at 4,s,20	Add
TIR 2,02	TIR_2_02	APH 1+2 outlet air, moves all 39 days	Add
IIR 5,1,3 · 6,1,1	pi_IIR_5_1_3/6_1_1_ActVal	ERH and EAOH3 heater currents	Add
String min. voltage	String_s_ActMinVoltage	Lowest stack voltage per string	Add
Min-cell limit	…ctEIRC4s_rVoltCellMin_Setp	Limit lowered 0.80 → 0.72 V	Add
Room climate	pi_QIR_10_23/24/25	16–25 °C, ~24 % RH, ~1000 mbar: may be the missing ambient reference	Check
Heater set-points	ct_TIRC_…_SP_TargetValue	Operator target temperatures	Optional
Compressor pressure	PT02_Haug	NG booster 0–16 bar	Optional

What to do next

1. Ask the plant team to reconnect or rescale the package-voltage channels; until then drop them from the screen. 2. Replace TIR 4,6,19; check the mounting of TIR 4,6,25. 3. Delete or fix rows 55 and 64. 4. Add EIR 4,s,21 and TIR 2,02; decide on the room-climate sensors. 5. Confirm EIRC 4,s,3/8/9 and EIRC 3,1,1/5,1,x against the ICA list.

If you remember one thing: Quickest wins: add the stack-4 voltages (EIR 4,s,21) and remove the 40 empty package-voltage channels.

5 · All 255 rows

#	Tag	Verdict	Logged as	Reason	Real %	Days
1	PIR 1,1,1	Low value	PIR_1_1_1	supply line pressure: almost constant (4-6 bar); only useful to spot a supply fault	99.0	33.0
2	FIRC 1,1,1	Key	ct_FIRC_111_PV_ProcessValue	Cathode air flow 1 - plant on/off and cooling rate	99.0	26.0
3	TIR 1,1,1	Low value	TIR_1_1_1	near room temperature most of the time; little change between stages (part of the ambient proxy in the stage rules)	99.0	39.0
4	FIRC 1,1,3	Limited	ct_FIRC_113_PV_ProcessValue	real values only from 2026-03-17 (tag renamed in PLC); shows hot-standby vs. cold clearly when present	29.1	12.0
5	PIR 1,1,2	Useful	PIR_1_1_2	working, follows the plant through the stages	99.0	33.0
6	PIR 1,1,3	Back-up twin	PIR_1_1_3	line-2 partner of PIR 1,1,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	33.0
7	FIRC 1,1,2	Key	ct_FIRC_112_PV_ProcessValue	Cathode air flow 2	99.0	26.0
8	PIRZ 8,1,2	Key	PIRZ_8_1_2	Safety pressure monitor	99.0	39.0
9	QIR 1,4	Not usable	PLC_pi_QIR_1_4_ActVal_Value	gas-quality value ≈0 nearly all the time; 4 short spikes only	99.0	6.0
10	PIR 1,4,1	Low value	PIR_1_4_1	supply line pressure: almost constant (4-6 bar); only useful to spot a supply fault	99.0	23.0
11	FIRC 1,4	Key	ct_FIRC_14_PV_ProcessValue	Natural gas - fuel phase	99.0	0.0
12	PIR 1,4,2	Useful	PIR_1_4_2	working, follows the plant through the stages	99.0	39.0
13	TIR 1,4,1	Low value	TIR_1_4_1	near room temperature most of the time; little change between stages (part of the ambient proxy in the stage rules)	99.0	39.0
14	PIR 1,2,1	Low value	PIR_1_2_1	supply line pressure: almost constant (4-6 bar); only useful to spot a supply fault	99.0	39.0
15	PIR 1,5	Low value	PIR_1_5	supply line pressure: almost constant (4-6 bar); only useful to spot a supply fault	99.0	39.0
16	PIR 1,3,1	Low value	PIR_1_3_1	supply line pressure: almost constant (4-6 bar); only useful to spot a supply fault	99.0	39.0
17	FIRC 1,2	Key	ct_FIRC_12_PV_ProcessValue	POx air - only with natural gas	99.0	0.0
18	FIRC 1,3	Key	ct_FIRC_13_PV_ProcessValue	Nitrogen - forming-gas / purge phase	99.0	22.0
19	FIRC 1,5	Key	ct_FIRC_15_PV_ProcessValue	Hydrogen - forming-gas phase	99.0	22.0
20	PIR 1,2,2	Useful	PIR_1_2_2	working, follows the plant through the stages	99.0	36.0
21	PIR 1,3,2	Useful	PIR_1_3_2	working, follows the plant through the stages	99.0	30.0
22	TIR 1,2,1	Low value	TIR_1_2_1	near room temperature most of the time; little change between stages (part of the ambient proxy in the stage rules)	99.0	39.0
23	TIR 1,2,2	Low value	TIR_1_2_2	near room temperature most of the time; little change between stages	99.0	39.0
24	TIR 1,3	Low value	TIR_1_3	near room temperature most of the time; little change between stages (part of the ambient proxy in the stage rules)	99.0	39.0
25	TIR 1,4,2	Back-up twin	TIR_1_4_2	line-2 partner of TIR 1,2,2 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	39.0
26	TIR 2,05	Key	TIR_2_05	Air entering the EAPH heater	99.0	36.0
27	PIR 2,01	Useful	PIR_2_01	working, follows the plant through the stages	99.0	33.0
28	TIR 2,03	Key	TIR_2_03	APH outlet - air pre-heat	99.0	35.0
29	EIRC 2,01	Key	EIRC_2_01_ActV_In_ThreePhasePower	EAPH heater power - shows when the plant is heating	99.0	32.0
30	TIRZ 8,1,1	Back-up twin	TIRZ_8_1_1	safety/control twin of TIRC 2,06 - same spot, moves 1:1 (r ≥ 0.96); keep for trip checks, drop from models	99.0	35.0
31	TIR 3,01	Low value	TIR_3_01	near room temperature most of the time; little change between stages	99.0	39.0
32	TIR 7,04	Useful	TIR_7_04	working, follows the plant through the stages	99.0	36.0
33	PIR 7,02	Useful	PIR_7_02	working, follows the plant through the stages	99.0	39.0
34	LIR 7,01	Not traceable	—	no column with this tag in any of the 836 logged columns		
35	TIR 2,04	Low value	TIR_2_04	near room temperature most of the time; little change between stages	99.0	39.0
36	TIR 3,04	Low value	TIR_3_04	near room temperature most of the time; little change between stages	99.0	39.0

#	Tag	Verdict	Logged as	Reason	Real %	Days
37	TIR 7,05	Back-up twin	TIR_7_05	safety/control twin of TIRZ 8,1,10 - same spot, moves 1:1 (r ≥ 0.96); keep for trip checks, drop from models	99.0	36.0
38	TIRZ 8,1,10	Key	TIRZ_8_1_10	Exhaust trip sensor (8 of 10 part-load runs tripped on it)	99.0	37.0
39	PIR 7,03	Useful	PIR_7_03	working, follows the plant through the stages	99.0	39.0
40	TIR 2,1,1	Low value	TIR_2_1_1	near room temperature most of the time; little change between stages	99.0	39.0
41	PIR 7,1,1	Useful	PIR_7_1_1	working, follows the plant through the stages	99.0	38.0
42	TIR 7,2,1	Useful	TIR_7_2_1	working, follows the plant through the stages	99.0	36.0
43	TIR 5,6,1	Useful	TIR_5_6_1	working, follows the plant through the stages	99.0	37.0
44	TIR 7,1,1	Useful	TIR_7_1_1	working, follows the plant through the stages	99.0	36.0
45	TIRC 3,17	Useful	TIRC_3_17	working, follows the plant through the stages	99.0	37.0
46	TIR 7,3,1	Useful	TIR_7_3_1	working, follows the plant through the stages	99.0	37.0
47	TIR 7,3,2	Back-up twin	TIR_7_3_2	line-2 partner of TIR 7,3,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	38.0
48	TIR 2,1,2	Back-up twin	TIR_2_1_2	line-2 partner of TIR 2,1,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	39.0
49	PIR 7,1,2	Limited	PIR_7_1_2	reads -1 (not connected) until 2026-02-17	99.0	20.0
50	TIR 7,2,2	Back-up twin	TIR_7_2_2	line-2 partner of TIR 7,2,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	37.0
51	TIR 7,1,2	Back-up twin	TIR_7_1_2	line-2 partner of TIR 7,1,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	37.0
52	TIR 5,6,2	Back-up twin	TIR_5_6_2	line-2 partner of TIR 5,6,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	37.0
53	TIRZ 8,2,5	Back-up twin	TIRZ_8_2_5	safety/control twin of TIRC 5,17 - same spot, moves 1:1 (r ≥ 0.96); keep for trip checks, drop from models	99.0	38.0
54	TIRC 5,17	Key	TIRC_5_17	After-burner housing heater (ERH) temperature	99.0	36.0
55	TIRC 3,17	Duplicate row	TIRC_3_17	same tag already listed as row 45	99.0	37.0
56	TIR 3,06	Useful	TIR_3_06	working, follows the plant through the stages	99.0	38.0
57	TIR 3,05	Low value	TIR_3_05	near room temperature most of the time; little change between stages	99.0	39.0
58	TIRC 3,03	Useful	TIRC_3_03	working, follows the plant through the stages	99.0	35.0
59	PIR 3,02	Useful	PIR_3_02	working, follows the plant through the stages	99.0	36.0
60	TIR 3,02	Low value	TIR_3_02	near room temperature most of the time; little change between stages	99.0	39.0
61	PIR 3,01	Useful	PIR_3_01	working, follows the plant through the stages	99.0	39.0
62	TIRZ 8,2,4	Back-up twin	TIRZ_8_2_4	safety/control twin of TIRC 6,1,1 - same spot, moves 1:1 (r ≥ 0.96); keep for trip checks, drop from models	99.0	39.0
63	TIRC 6,1,1	Key	TIRC_6_1_1	EAOH3 heater temperature	99.0	39.0
64	EIRC 2,01	Duplicate row	EIRC_2_01_ActV_In_ThreePhasePower	same tag already listed as row 29	99.0	32.0
65	TIR 5,4,2	Back-up twin	TIR_5_4_2	line-2 partner of TIR 5,4,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	36.0
66	PIR 5,3,2	Back-up twin	PIR_5_3_2	line-2 partner of PIR 5,3,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	35.0
67	TIR 6,01	Useful	TIR_6_01	working, follows the plant through the stages	99.0	38.0
68	PIR 6,01	Useful	PIR_6_01	working, follows the plant through the stages	99.0	35.0
69	TIR 5,4,1	Useful	TIR_5_4_1	working, follows the plant through the stages	99.0	36.0
70	PIR 5,3,1	Useful	PIR_5_3_1	working, follows the plant through the stages	99.0	35.0
71	TIRC 5,5,1	Useful	TIRC_5_5_1	working, follows the plant through the stages	99.0	36.0
72	PIR 5,4,1	Useful	PIR_5_4_1	working, follows the plant through the stages	99.0	35.0
73	TIRC 5,5,2	Back-up twin	TIRC_5_5_2	line-2 partner of TIRC 5,5,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	36.0
74	PIR 5,4,2	Back-up twin	PIR_5_4_2	line-2 partner of PIR 5,4,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	35.0
75	TIRZ 8,2,2	Back-up twin	TIRZ_8_2_2	safety/control twin of TIRC 5,3,1 - same spot, moves 1:1 (r ≥ 0.96); keep for trip checks, drop from models	99.0	37.0

#	Tag	Verdict	Logged as	Reason	Real %	Days
76	TIRC 5,3,1	Key	TIRC_5_3_1	EAOH1 heater temperature	99.0	36.0
77	EIRC 5,1,1	Useful	pi_IIR_5_1_1_ActVal	heater current (logged as IIR) - shows when the heater is on | match to data column is assumed	99.0	31.0
78	TIRZ 8,2,3	Back-up twin	TIRZ_8_2_3	safety/control twin of TIRC 5,3,2 - same spot, moves 1:1 (r ≥ 0.96); keep for trip checks, drop from models	99.0	37.0
79	TIRC 5,3,2	Key	TIRC_5_3_2	EAOH2 heater temperature	99.0	37.0
80	EIRC 5,1,2	Useful	pi_IIR_5_1_2_ActVal	heater current (logged as IIR) - shows when the heater is on | match to data column is assumed	99.0	31.0
81	TIRZ 8,1,9	Useful	TIRZ_8_1_9	working, follows the plant through the stages	99.0	37.0
82	TIRC 5,07	Useful	TIRC_5_07	working, follows the plant through the stages	99.0	37.0
83	TIRC 5,08	Useful	TIRC_5_08	working, follows the plant through the stages	99.0	36.0
84	TIRC 5,09	Key	TIRC_5_09	After-burner (TOX) representative temperature	99.0	37.0
85	TIRC 5,10	Useful	TIRC_5_10	working, follows the plant through the stages	99.0	36.0
86	TIRC 5,11	Useful	TIRC_5_11	working, follows the plant through the stages	99.0	36.0
87	TIRC 5,12	Useful	TIRC_5_12	working, follows the plant through the stages	99.0	37.0
88	TIRC 5,13	Useful	TIRC_5_13	working, follows the plant through the stages	99.0	37.0
89	TIRC 5,14	Useful	TIRC_5_14	working, follows the plant through the stages	99.0	37.0
90	TIRC 5,15	Useful	TIRC_5_15	working, follows the plant through the stages	99.0	37.0
91	TIRC 3,07	Key	TIRC_3_07	Reformer catalyst inlet	99.0	36.0
92	TIRC 3,08	Useful	TIRC_3_08	working, follows the plant through the stages	99.0	36.0
93	TIRC 3,09	Useful	TIRC_3_09	working, follows the plant through the stages	99.0	35.0
94	TIRC 3,10	Useful	TIRC_3_10	working, follows the plant through the stages	99.0	36.0
95	TIRC 3,11	Useful	TIRC_3_11	working, follows the plant through the stages	99.0	36.0
96	TIRC 3,12	Useful	TIRC_3_12	working, follows the plant through the stages	99.0	36.0
97	TIRC 3,13	Useful	TIRC_3_13	working, follows the plant through the stages	99.0	37.0
98	TIRC 3,14	Useful	TIRC_3_14	working, follows the plant through the stages	99.0	36.0
99	TIRC 3,15	Key	TIRC_3_15	Reformer catalyst outlet	99.0	37.0
100	TIRC 3,16	Useful	TIRC_3_16	working, follows the plant through the stages	99.0	36.0
101	EIRC 3,1,1	Useful	pi_IIR_3_1_1_ActVal	heater current (logged as IIR) - shows when the heater is on | match to data column is assumed	99.0	31.0
102	TIRC 3,1,1	Key	TIRC_3_1_1	EAGH heater temperature	99.0	37.0
103	TIRZ 8,2,1	Back-up twin	TIRZ_8_2_1	safety/control twin of TIRC 3,1,1 - same spot, moves 1:1 (r ≥ 0.96); keep for trip checks, drop from models	99.0	37.0
104	PIR 3,03	Useful	PIR_3_03	working, follows the plant through the stages	99.0	36.0
105	PIR 2,02	Key	PIR_2_02	Cathode inlet pressure (used in stage rules)	99.0	33.0
106	TIR 3,18	Useful	TIR_3_18	working, follows the plant through the stages	99.0	37.0
107	TIRC 2,06	Key	TIRC_2_06	EAPH heater control temperature (start-up ramp)	99.0	33.0
108	TIRC 3,19,2	Back-up twin	TIR_3_19_2	line-2 partner of TIRC 3,19,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	37.0
109	TIRC 3,19,1	Useful	TIR_3_19_1	working, follows the plant through the stages	99.0	37.0
110	TIR 4,6,1	Key (as a group)	TIR_4_6_1	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	34.0
111	TIR 4,6,2	Key (as a group)	TIR_4_6_2	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	34.0
112	TIR 4,6,3	Key (as a group)	TIR_4_6_3	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	34.0

#	Tag	Verdict	Logged as	Reason	Real %	Days
113	TIR 4,6,4	Key (as a group)	TIR_4_6_4	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	34.0
114	TIR 4,6,5	Key (as a group)	TIR_4_6_5	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	34.0
115	TIR 4,6,6	Key (as a group)	TIR_4_6_6	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	34.0
116	TIR 4,6,7	Key (as a group)	TIR_4_6_7	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	34.0
117	TIR 4,6,8	Key (as a group)	TIR_4_6_8	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	34.0
118	TIR 4,6,9	Key (as a group)	TIR_4_6_9	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
119	TIR 4,6,10	Key (as a group)	TIR_4_6_10	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
120	TIR 4,6,11	Key (as a group)	TIR_4_6_11	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
121	TIR 4,6,12	Key (as a group)	TIR_4_6_12	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
122	TIR 4,6,13	Key (as a group)	TIR_4_6_13	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
123	TIR 4,6,14	Key (as a group)	TIR_4_6_14	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
124	TIR 4,6,15	Key (as a group)	TIR_4_6_15	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
125	TIR 4,6,16	Key (as a group)	TIR_4_6_16	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
126	TIR 4,6,17	Key (as a group)	TIR_4_6_17	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
127	TIR 4,6,18	Key (as a group)	TIR_4_6_18	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
128	TIR 4,6,19	Not usable	TIR_4_6_19	stuck at 1300 °C on all 39 days - broken thermocouple	99.0	0.0
129	TIR 4,6,20	Key (as a group)	TIR_4_6_20	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
130	TIR 4,6,21	Key (as a group)	TIR_4_6_21	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
131	TIR 4,6,22	Key (as a group)	TIR_4_6_22	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
132	TIR 4,6,23	Key (as a group)	TIR_4_6_23	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
133	TIR 4,6,24	Key (as a group)	TIR_4_6_24	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
134	TIR 4,6,25	Not usable	TIR_4_6_25	reads 2-66 °C while the other stack sensors read 500-650 °C - not in the stack (wrong spot or loose)	99.0	39.0
135	TIR 4,6,26	Key (as a group)	TIR_4_6_26	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0

#	Tag	Verdict	Logged as	Reason	Real %	Days
136	TIR 4,6,27	Key (as a group)	TIR_4_6_27	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
137	TIR 4,6,28	Key (as a group)	TIR_4_6_28	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	36.0
138	TIR 4,6,29	Key (as a group)	TIR_4_6_29	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
139	TIR 4,6,30	Key (as a group)	TIR_4_6_30	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
140	TIR 4,6,31	Key (as a group)	TIR_4_6_31	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
141	TIR 4,6,32	Key (as a group)	TIR_4_6_32	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
142	TIR 4,6,33	Key (as a group)	TIR_4_6_33	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
143	TIR 4,6,34	Key (as a group)	TIR_4_6_34	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
144	TIR 4,6,35	Key (as a group)	TIR_4_6_35	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
145	TIR 4,6,36	Key (as a group)	TIR_4_6_36	stack temperature grid: one sensor alone repeats the others (r = 1.00), but together they give the stack mean, hottest spot and spread	99.0	35.0
146	TIRZ 8,1,3	Key	TIRZ_8_1_3	Stack safety temperature	99.0	35.0
147	TIRC 5,1,2	Back-up twin	TIRC_5_1_2	line-2 partner of TIRC 5,1,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	36.0
148	PIR 5,1,2	Back-up twin	PIR_5_1_2	line-2 partner of PIR 5,1,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	35.0
149	TIRC 5,1,1	Key	TIRC_5_1_1	Stack cathode outlet 1	99.0	35.0
150	PIR 5,1,1	Useful	PIR_5_1_1	working, follows the plant through the stages	99.0	35.0
151	TIR 4,1	Low value	TIR_4_1	near room temperature most of the time; little change between stages	99.0	39.0
152	EIR 4,1,10	Not usable	pi_EIR_4_1_10	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
153	EIR 4,1,11	Not usable	pi_EIR_4_1_11	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	6.0
154	EIR 4,1,12	Key	pi_EIR_4_1_12	stack 1 voltage of string 1 (≈110 V hot, 0 V cold)	99.0	29.0
155	EIR 4,1,13	Not usable	pi_EIR_4_1_13	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
156	EIR 4,1,14	Not usable	pi_EIR_4_1_14	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	6.0
157	EIR 4,1,15	Key	pi_EIR_4_1_15	stack 2 voltage of string 1 (≈110 V hot, 0 V cold)	99.0	28.0
158	EIR 4,1,16	Not usable	pi_EIR_4_1_16	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
159	EIR 4,1,17	Not usable	pi_EIR_4_1_17	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
160	EIR 4,1,18	Key	pi_EIR_4_1_18	stack 3 voltage of string 1 (≈110 V hot, 0 V cold)	99.0	30.0
161	EIR 4,1,19	Not usable	pi_EIR_4_1_19	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
162	EIR 4,1,20	Not usable	pi_EIR_4_1_20	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0

#	Tag	Verdict	Logged as	Reason	Real %	Days
163	EIR 4,2,10	Not usable	pi_EIR_4_2_10	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
164	EIR 4,2,11	Not usable	pi_EIR_4_2_11	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
165	EIR 4,2,12	Key	pi_EIR_4_2_12	stack 1 voltage of string 2 (≈110 V hot, 0 V cold)	99.0	31.0
166	EIR 4,2,13	Not usable	pi_EIR_4_2_13	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
167	EIR 4,2,14	Not usable	pi_EIR_4_2_14	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
168	EIR 4,2,15	Key	pi_EIR_4_2_15	stack 2 voltage of string 2 (≈110 V hot, 0 V cold)	99.0	30.0
169	EIR 4,2,16	Not usable	pi_EIR_4_2_16	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
170	EIR 4,2,17	Not usable	pi_EIR_4_2_17	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
171	EIR 4,2,18	Key	pi_EIR_4_2_18	stack 3 voltage of string 2 (≈110 V hot, 0 V cold)	99.0	32.0
172	EIR 4,2,19	Not usable	pi_EIR_4_2_19	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
173	EIR 4,2,20	Not usable	pi_EIR_4_2_20	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
174	EIR 4,3,10	Not usable	pi_EIR_4_3_10	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
175	EIR 4,3,11	Not usable	pi_EIR_4_3_11	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
176	EIR 4,3,12	Key	pi_EIR_4_3_12	stack 1 voltage of string 3 (≈110 V hot, 0 V cold)	99.0	31.0
177	EIR 4,3,13	Not usable	pi_EIR_4_3_13	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
178	EIR 4,3,14	Not usable	pi_EIR_4_3_14	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
179	EIR 4,3,15	Key	pi_EIR_4_3_15	stack 2 voltage of string 3 (≈110 V hot, 0 V cold)	99.0	32.0
180	EIR 4,3,16	Not usable	pi_EIR_4_3_16	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
181	EIR 4,3,17	Not usable	pi_EIR_4_3_17	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
182	EIR 4,3,18	Key	pi_EIR_4_3_18	stack 3 voltage of string 3 (≈110 V hot, 0 V cold)	99.0	27.0
183	EIR 4,3,19	Not usable	pi_EIR_4_3_19	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
184	EIR 4,3,20	Not usable	pi_EIR_4_3_20	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
185	EIR 4,4,10	Not usable	pi_EIR_4_4_10	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
186	EIR 4,4,11	Not usable	pi_EIR_4_4_11	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
187	EIR 4,4,12	Key	pi_EIR_4_4_12	stack 1 voltage of string 4 (≈110 V hot, 0 V cold)	99.0	32.0
188	EIR 4,4,13	Not usable	pi_EIR_4_4_13	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
189	EIR 4,4,14	Not usable	pi_EIR_4_4_14	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
190	EIR 4,4,15	Key	pi_EIR_4_4_15	stack 2 voltage of string 4 (≈110 V hot, 0 V cold)	99.0	31.0
191	EIR 4,4,16	Not usable	pi_EIR_4_4_16	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	36.0
192	EIR 4,4,17	Not usable	pi_EIR_4_4_17	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	35.0
193	EIR 4,4,18	Key	pi_EIR_4_4_18	stack 3 voltage of string 4 (≈110 V hot, 0 V cold)	99.0	29.0

#	Tag	Verdict	Logged as	Reason	Real %	Days
194	EIR 4,4,19	Not usable	pi_EIR_4_4_19	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
195	EIR 4,4,20	Not usable	pi_EIR_4_4_20	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
196	EIR 4,5,10	Not usable	pi_EIR_4_5_10	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
197	EIR 4,5,11	Not usable	pi_EIR_4_5_11	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
198	EIR 4,5,12	Key	pi_EIR_4_5_12	stack 1 voltage of string 5 (≈110 V hot, 0 V cold)	99.0	28.0
199	EIR 4,5,13	Not usable	pi_EIR_4_5_13	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
200	EIR 4,5,14	Not usable	pi_EIR_4_5_14	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
201	EIR 4,5,15	Key	pi_EIR_4_5_15	stack 2 voltage of string 5 (≈110 V hot, 0 V cold)	99.0	28.0
202	EIR 4,5,16	Not usable	pi_EIR_4_5_16	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
203	EIR 4,5,17	Not usable	pi_EIR_4_5_17	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
204	EIR 4,5,18	Key	pi_EIR_4_5_18	stack 3 voltage of string 5 (≈110 V hot, 0 V cold)	99.0	29.0
205	EIR 4,5,19	Not usable	pi_EIR_4_5_19	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
206	EIR 4,5,20	Not usable	pi_EIR_4_5_20	package voltage: real readings (≈5 V) only on 13-15 Nov 2025, ≈0 V since 16 Nov 2025 - channel lost or unscaled	99.0	5.0
207	EIRC 4,1,1	Key	PLC_hostSetp_ctrl_ctEIRC41_rVoltMonitor	string load voltage (only non-zero while drawing current)	99.0	6.0
208	EIRC 4,1,2	Key	PLC_hostSetp_ctrl_ctEIRC41_rCurrMonitor	string current - the only measure of electrical load	99.0	7.0
209	EIRC 4,1,3	Limited	EIRC_41_Actual_Power	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20 | match to data column is assumed	27.9	4.0
210	EIRC 4,1,4	Useful	PLC_hostSetp_ctrl_ctEIRC41_SP_TargetValue_Value	current set-point given to the load	99.0	8.0
211	EIRC 4,1,5	Low value	PLC_hostSetp_ctrl_ctEIRC41_SP_SlewRateAbs	slew-rate setting: stays at one value - a setting, not a measurement	99.0	7.0
212	EIRC 4,1,6	Low value	PLC_hostSetp_ctrl_ctEIRC41_OpMode	operating-mode setting (0 or 3); same information as load on/off	99.0	6.0
213	EIRC 4,1,7	Limited	EIRC_41_OUT_Enable	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20	27.9	4.0
214	EIRC 4,1,8	Limited	EIRC_41_OUT_StackCon_OnOff)	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20 | match to data column is assumed	29.1	1.0
215	EIRC 4,1,9	Key	PLC_hostSetp_ctrl_ctEIRC41_rVoltCellMin_ActV	lowest voltage seen by the load - the limit that capped part load | match to data column is assumed	99.0	21.0
216	EIRC 4,2,1	Key	PLC_hostSetp_ctrl_ctEIRC42_rVoltMonitor	string load voltage (only non-zero while drawing current)	99.0	36.0
217	EIRC 4,2,2	Key	PLC_hostSetp_ctrl_ctEIRC42_rCurrMonitor	string current - the only measure of electrical load	99.0	6.0
218	EIRC 4,2,3	Limited	EIRC_42_Actual_Power	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20 | match to data column is assumed	29.1	4.0
219	EIRC 4,2,4	Useful	PLC_hostSetp_ctrl_ctEIRC42_SP_TargetValue_Value	current set-point given to the load	99.0	8.0
220	EIRC 4,2,5	Low value	PLC_hostSetp_ctrl_ctEIRC42_SP_SlewRateAbs	slew-rate setting: stays at one value - a setting, not a measurement	99.0	6.0
221	EIRC 4,2,6	Low value	PLC_hostSetp_ctrl_ctEIRC42_OpMode	operating-mode setting (0 or 3); same information as load on/off	99.0	0.0
222	EIRC 4,2,7	Limited	EIRC_42_OUT_Enable	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20	27.9	4.0

#	Tag	Verdict	Logged as	Reason	Real %	Days
223	EIRC 4,2,8	Limited	EIRC_42_OUT_StackCon_OnOff)	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20 | match to data column is assumed	29.1	4.0
224	EIRC 4,2,9	Key	PLC_hostSetp_ctrl_ctEIRC42_rVoltCellMin_ActV	lowest voltage seen by the load - the limit that capped part load | match to data column is assumed	99.0	21.0
225	EIRC 4,3,1	Key	PLC_hostSetp_ctrl_ctEIRC43_rVoltMonitor	string load voltage (only non-zero while drawing current)	99.0	6.0
226	EIRC 4,3,2	Key	PLC_hostSetp_ctrl_ctEIRC43_rCurrMonitor	string current - the only measure of electrical load	99.0	7.0
227	EIRC 4,3,3	Limited	EIRC_43_Actual_Power	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20 | match to data column is assumed	29.1	5.0
228	EIRC 4,3,4	Useful	PLC_hostSetp_ctrl_ctEIRC43_SP_TargetValue_Value	current set-point given to the load	99.0	8.0
229	EIRC 4,3,5	Low value	PLC_hostSetp_ctrl_ctEIRC43_SP_SlewRateAbs	slew-rate setting: stays at one value - a setting, not a measurement	99.0	7.0
230	EIRC 4,3,6	Low value	PLC_hostSetp_ctrl_ctEIRC43_OpMode	operating-mode setting (0 or 3); same information as load on/off	99.0	0.0
231	EIRC 4,3,7	Limited	EIRC_43_OUT_Enable	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20	29.1	5.0
232	EIRC 4,3,8	Limited	EIRC_43_OUT_StackCon_OnOff)	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20 | match to data column is assumed	29.1	6.0
233	EIRC 4,3,9	Key	PLC_hostSetp_ctrl_ctEIRC43_rVoltCellMin_ActV	lowest voltage seen by the load - the limit that capped part load | match to data column is assumed	99.0	21.0
234	EIRC 4,4,1	Key	PLC_hostSetp_ctrl_ctEIRC44_rVoltMonitor	string load voltage (only non-zero while drawing current)	99.0	0.0
235	EIRC 4,4,2	Key	PLC_hostSetp_ctrl_ctEIRC44_rCurrMonitor	string current - the only measure of electrical load	99.0	6.0
236	EIRC 4,4,3	Limited	EIRC_44_Actual_Power	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20 | match to data column is assumed	29.1	4.0
237	EIRC 4,4,4	Useful	PLC_hostSetp_ctrl_ctEIRC44_SP_TargetValue_Value	current set-point given to the load	99.0	8.0
238	EIRC 4,4,5	Low value	PLC_hostSetp_ctrl_ctEIRC44_SP_SlewRateAbs	slew-rate setting: stays at one value - a setting, not a measurement	99.0	6.0
239	EIRC 4,4,6	Low value	PLC_hostSetp_ctrl_ctEIRC44_OpMode	operating-mode setting (0 or 3); same information as load on/off	99.0	0.0
240	EIRC 4,4,7	Limited	EIRC_44_OUT_Enable	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20	29.1	4.0
241	EIRC 4,4,8	Limited	EIRC_44_OUT_StackCon_OnOff)	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20 | match to data column is assumed	29.1	5.0
242	EIRC 4,4,9	Key	PLC_hostSetp_ctrl_ctEIRC44_rVoltCellMin_ActV	lowest voltage seen by the load - the limit that capped part load | match to data column is assumed	99.0	19.0
243	EIRC 4,5,1	Key	PLC_hostSetp_ctrl_ctEIRC45_rVoltMonitor	string load voltage (only non-zero while drawing current)	99.0	0.0
244	EIRC 4,5,2	Key	PLC_hostSetp_ctrl_ctEIRC45_rCurrMonitor	string current - the only measure of electrical load	99.0	6.0
245	EIRC 4,5,3	Limited	EIRC_45_Actual_Power	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20 | match to data column is assumed	29.1	4.0
246	EIRC 4,5,4	Useful	PLC_hostSetp_ctrl_ctEIRC45_SP_TargetValue_Value	current set-point given to the load	99.0	8.0
247	EIRC 4,5,5	Low value	PLC_hostSetp_ctrl_ctEIRC45_SP_SlewRateAbs	slew-rate setting: stays at one value - a setting, not a measurement	99.0	7.0
248	EIRC 4,5,6	Low value	PLC_hostSetp_ctrl_ctEIRC45_OpMode	operating-mode setting (0 or 3); same information as load on/off	99.0	0.0
249	EIRC 4,5,7	Limited	EIRC_45_OUT_Enable	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20	29.1	4.0
250	EIRC 4,5,8	Limited	EIRC_45_OUT_StackCon_OnOff)	only logged from 2026-03-17; device status column has shifted/garbage values on 2026-03-20 | match to data column is assumed	29.1	5.0

#	Tag	Verdict	Logged as	Reason	Real %	Days
251	EIRC 4,5,9	Key	PLC_hostSetp_ctrl_ctEIRC45_rVoltCellMin_ActV	lowest voltage seen by the load - the limit that capped part load | match to data column is assumed	97.8	21.0
252	TIRC 5,2,1	Key	TIRC_5_2_1	Stack anode outlet 1	99.0	37.0
253	TIRC 5,2,2	Back-up twin	TIRC_5_2_2	line-2 partner of TIRC 5,2,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	36.0
254	PIR 5,2,1	Useful	PIR_5_2_1	working, follows the plant through the stages	99.0	37.0
255	PIR 5,2,2	Back-up twin	PIR_5_2_2	line-2 partner of PIR 5,2,1 - moves 1:1 (r ≥ 0.98); keep only to compare the two lines	99.0	38.0